

ISTRANA, Italy

WMO: 160980

Lat: 45.685N

Lon: 12.083E

Elev: 137

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	26.2	9.9	9.2	35.1	15.4	12.0	34.7	19.0	41.4	15.7	41.2	3.0	40	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	93.4	74.5	90.8	73.2	88.1	71.8	77.8	86.8	75.9	85.2	74.3	83.8	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	133.2	82.4	73.3	124.4	80.8	71.4	116.5	79.3	41.3	86.8	39.4	84.5	37.9	83.6	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	13.7	11.6	DB	19.1	96.9	4.1	2.7	16.1	98.8	13.7	100.3	11.4	101.8	8.4	103.8	(4)
			WB	17.9	80.6	4.1	2.6	15.0	82.4	12.6	84.0	10.2	85.4	7.2	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	38.7	41.4	49.1	57.0	65.7	73.1	76.9	76.3	67.6	58.5	48.5	40.2	(6)
		DBStd	14.63	4.97	5.26	5.58	5.71	5.54	5.60	4.49	5.30	5.34	5.84	6.26	5.72	(7)
		HDD50	1093	350	242	82	9	0	0	0	0	7	98	305		(8)
		HDD65	3788	814	661	494	245	59	6	0	1	34	212	494	768	(9)
		CDD50	3958	1	2	53	219	486	693	835	815	527	270	55	2	(10)
		CDD65	1178	0	0	0	5	80	250	370	351	111	11	0	0	(11)
		CDH74	10732	0	0	1	62	666	2278	3665	3346	690	24	0	0	(12)
		CDH80	4147	0	0	0	5	161	847	1544	1437	152	1	0	0	(13)
(14)	Wind	WSAvg	4.3	3.7	4.4	4.9	5.3	4.9	4.7	4.5	4.2	4.1	3.7	3.8	3.5	(14)
(15)	Precipitation	PrecAvg	37.7	2.0	2.0	2.4	3.4	3.9	3.6	3.1	3.0	3.7	4.0	4.0	2.6	(15)
(16)		PrecMax	65.9	10.2	9.4	10.6	5.8	8.8	7.2	8.2	6.1	7.1	7.6	8.2	6.4	(16)
(17)		PrecMin	26.0	0.0	0.0	0.0	0.1	1.1	1.1	1.6	0.7	1.1	0.3	0.9	0.2	(17)
(18)		PrecStd	9.6	2.1	2.4	2.3	1.5	2.1	1.5	1.4	1.5	1.9	2.1	1.9	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.6	60.6	70.9	79.2	88.6	94.8	96.1	97.7	87.5	76.9	65.2	56.1		(19)
(20)		0.4% MCWB	49.3	48.6	54.6	62.0	68.4	74.1	74.4	74.1	69.3	64.8	58.8	51.5		(20)
(21)		2% DB	51.6	56.4	66.4	75.4	84.3	91.1	92.9	94.3	84.3	73.4	62.1	53.6		(21)
(22)		2% MCWB	46.4	46.6	52.8	59.2	66.7	73.4	75.2	74.0	68.6	63.7	57.0	49.2		(22)
(23)		5% DB	49.4	53.4	62.8	72.0	80.8	88.0	90.0	91.0	81.0	70.3	59.8	51.0		(23)
(24)		5% MCWB	44.9	45.5	51.6	58.1	64.9	71.6	73.2	73.3	67.2	62.2	55.6	46.9		(24)
(25)		10% DB	46.8	50.4	59.4	68.4	77.3	85.4	87.9	87.8	78.1	67.8	57.6	48.6		(25)
(26)		10% MCWB	43.1	44.8	50.1	56.7	63.5	70.4	71.9	72.1	65.8	60.9	53.8	45.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	51.6	51.2	57.9	64.4	72.8	79.1	80.4	80.9	73.8	68.5	61.2	53.7		(27)
(28)		0.4% MCDB	53.7	56.7	65.4	74.1	81.5	88.5	89.4	89.5	81.4	72.1	63.9	55.1		(28)
(29)		2% WB	48.2	49.2	55.2	61.6	69.7	75.9	77.9	77.7	71.6	66.4	58.6	50.3		(29)
(30)		2% MCDB	50.1	53.3	61.7	71.5	79.4	85.9	87.3	86.1	79.1	70.4	60.8	52.1		(30)
(31)		5% WB	45.8	47.5	53.5	59.7	67.3	73.8	76.0	75.8	69.9	64.4	56.7	47.8		(31)
(32)		5% MCDB	47.9	51.2	59.9	68.8	76.9	83.7	85.1	84.9	76.6	68.4	59.0	49.7		(32)
(33)		10% WB	43.7	45.8	51.9	58.0	65.4	72.0	74.3	74.1	68.0	62.4	54.6	46.0		(33)
(34)		10% MCDB	45.9	49.2	57.6	66.3	74.4	81.7	83.6	83.7	75.2	66.1	57.0	48.0		(34)
(35)	Mean Daily Temperature Range	MDBR	13.9	15.8	17.7	18.3	18.8	18.9	19.4	19.3	17.8	15.0	12.9	13.5		(35)
(36)		5% DB MCDBR	16.7	20.9	23.3	23.6	23.4	21.7	21.4	22.2	20.8	17.4	13.9	15.2		(36)
(37)		5% DB MCWBR	11.7	12.6	12.4	10.7	9.5	8.5	8.1	8.0	8.4	9.3	9.1	10.6		(37)
(38)		5% WB MCDBR	12.1	15.6	19.0	20.3	19.9	19.9	19.9	20.2	17.7	13.8	10.3	11.1		(38)
(39)		5% WB MCWBR	9.7	10.7	11.4	10.5	9.7	9.5	9.1	8.9	8.9	9.0	8.0	9.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.372	0.420	0.464	0.470	0.492	0.475	0.465	0.447	0.428	0.378	0.340		(40)
(41)		taud	2.388	2.294	2.168	2.075	2.112	2.101	2.157	2.199	2.211	2.219	2.312	2.417		(41)
(42)		Ebn at Noon	237	251	253	252	255	250	252	250	242	226	219	225		(42)
(43)		Edh at Noon	24	31	41	49	49	49	46	43	39	34	26	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	439	711	1117	1452	1772	2012	2048	1755	1286	797	463	380		(44)
(45)		RadStd	68	113	146	164	173	114	112	134	102	97	68	50		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.22	+1.73	+3.40	N/A	N/A	N/A	-167	-310	+211	N/A
Regional (53 neighbors)		+0.74	N/A	N/A	+1.08	+0.67	N/A	N/A	-209	+140	+49

Nomenclature: See separate page